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Siphoned Apart: A Portfolio Perspective on Order Flow Segmentation

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1 Big picture

- **Question.** What are the liquidity, profitability, and welfare consequences of order flow segmentation, especially payment for order flow (PFOF)?
- **Main contribution.** The paper gives an *inventory-based* theory of PFOF. It differs from the standard information-based view in which retail orders are attractive because they are less informed.
- **Core mechanism.** Market makers intermediate multiple order-flow sources. They choose a portfolio of liquidity supplies and trade off spread revenue against aggregate inventory risk. An order flow can be valuable not only because it is uninformed, but because it hedges other order flows.
- **Key idea.** Retail orders may be cheap to intermediate because their directionality is less correlated, or even negatively correlated, with institutional order flow. This makes them useful for market makers' inventory management.
- **Baseline model.** With several marketplaces and exogenous order-flow segmentation, market makers choose liquidity supplies across venues. Equilibrium spreads clear market-maker supply and investor demand.
- **Portfolio perspective.** A market maker's inventory cost depends on the covariance matrix of order-flow directionalities. Liquidity supply therefore resembles a Markowitz portfolio problem, even though the asset being traded is only one stock.
- **PFOF application.** The model specializes to retail (R) and institutional (I) order flow. Institutional orders trade on-exchange. Retail orders can be siphoned off-exchange by market makers.
- **Segmentation result.** PFOF can arise endogenously from inventory considerations: market makers siphon retail flow if it improves their inventory portfolio.
- **Segmentation ban.** A ban on off-exchange retail order flow always harms retail investors in the model. It can also harm institutional investors if retail flow is sufficiently useful for hedging institutional flow.
- **Market-maker profits.** PFOF can be a prisoner's dilemma: each market maker wants to siphon retail flow individually, but all market makers may be worse off once everyone does so.
- **Welfare result.** A segmentation ban reduces total welfare in the model because unconstrained market makers choose welfare-maximizing quantities of liquidity supply.
- **Takeaway.** The regulatory assessment of PFOF depends on the underlying friction. If segmentation is driven by inventory hedging rather than adverse selection, banning it can reduce welfare and raise trading costs.

2 Model primitives and notation

2.1 Marketplaces and order-flow directionality

There are J marketplaces. Each marketplace j has liquidity demand $\lambda_j(s_j)$, where s_j is the half bid-ask spread in that marketplace. Demand is decreasing in the spread:

$$\lambda'_j(s_j) < 0. \quad (1)$$

Liquidity-demanding orders in marketplace j have random *directionality* $D_j \in [-1, 1]$:

$$\Pr(\text{buy in } j) = \frac{1 + D_j}{2}, \quad \Pr(\text{sell in } j) = \frac{1 - D_j}{2}. \quad (2)$$

Let

$$\mathbf{D} = (D_1, D_2, \dots, D_J)', \quad \boldsymbol{\mu} = \mathbb{E}[\mathbf{D}], \quad \boldsymbol{\Sigma} = \text{var}(\mathbf{D}). \quad (3)$$

Interpretation: Directionality captures order-flow imbalance. Correlation across D_j captures whether different order-flow sources tend to buy and sell in the same direction or offset one another.

2.2 Liquidity demand microfoundation

The paper motivates downward-sloping liquidity demand with investors whose private values are random. If investor private-value magnitude has c.d.f. F_j , and if there is a mass κ_j of potential investors, then

$$\lambda_j(s_j) = \kappa_j [1 - F_j(s_j)]. \quad (4)$$

Interpretation: The spread screens orders: a larger spread discourages investors with lower private values from submitting marketable orders.

2.3 Market makers and liquidity supply

There is a continuum of market makers indexed by $m \in [0, M]$. Market maker m chooses liquidity supply

$$\mathbf{x}_m = (x_{m1}, x_{m2}, \dots, x_{mJ})'. \quad (5)$$

If she supplies x_{mj} in marketplace j , the number of orders she receives is Poisson with mean x_{mj} . Let Q_{mj} be her realized volume and Z_{mj} her realized net inventory in marketplace j .

Interpretation: The model uses quantity competition in liquidity supply. The spread is the price, and market makers choose how much liquidity to supply at that price.

2.4 Market-maker expected profit

A market maker earns spread revenue and bears quadratic inventory cost:

$$\mathbb{E} \left[\sum_{j=1}^J Q_{mj} s_j - \frac{\gamma}{2} \left(\sum_{j=1}^J Z_{mj} \right)^2 \right]. \quad (1)$$

Here $\gamma > 0$ scales inventory costs.

Interpretation: Inventory risk is not venue-by-venue. It is portfolio-wide: net inventory across all marketplaces determines the cost.

2.5 Equilibrium definition

An equilibrium consists of liquidity supplies $\mathbf{x}_m$ and spreads s_j such that:

1. each market maker chooses $\mathbf{x}_m$ to maximize expected profit; and
2. market clearing holds in each marketplace:

$$\int_0^M x_{mj} dm = \lambda_j(s_j), \quad j = 1, \dots, J. \quad (2)$$

Interpretation: Spreads adjust so that total market-maker liquidity supply equals investor liquidity demand.

3 Baseline model: portfolio-based liquidity supply

3.1 Net inventory and Lemma 1

The net inventory of market maker m in marketplace j is

$$Z_{mj} = \sum_{i=1}^{Q_{mj}} (-1)^{B_{imj}}, \quad (3)$$

where B_{imj} determines whether order i is a buy or sell.

Lemma 1. Market maker m 's expected squared inventory is a quadratic function of her liquidity-supply vector. Intuitively, it contains:

- a term reflecting the variance of directionality, $\mathbf{x}'_m \boldsymbol{\Sigma} \mathbf{x}_m$;
- a term reflecting average directionality, $(\boldsymbol{\mu}' \mathbf{x}_m)^2$; and
- terms reflecting the Poisson randomness of order arrivals.

Interpretation: This lemma converts liquidity provision into a portfolio problem. Expected inventory cost depends on the covariance matrix of order-flow directionalities.

3.2 Market maker objective

Using Lemma 1, each market maker's problem can be written as a quadratic optimization problem:

$$\max_{\mathbf{x}_m \geq 0} \left\{ \mathbf{x}'_m \mathbf{s} - \frac{\gamma}{2} \mathcal{I}(\mathbf{x}_m; \boldsymbol{\mu}, \boldsymbol{\Sigma}) \right\}, \quad (6)$$

where $\mathcal{I}(\cdot)$ is the inventory-risk term implied by Lemma 1.

Interpretation: The market maker chooses a portfolio of order flows. Spreads are expected returns; inventory risk is analogous to portfolio variance.

3.3 Proposition 1: equilibrium liquidity supply

Proposition 1. There exists a unique equilibrium. Equilibrium spreads and liquidity supplies solve a system in which each spread compensates market makers for the marginal inventory cost of supplying liquidity in that marketplace.

Interpretation: In equilibrium, a marketplace with riskier or more correlated order flow must offer a larger spread to induce market makers to supply liquidity.

3.4 Connection to portfolio theory

The market maker problem is analogous to a mean-variance portfolio problem. In portfolio theory, an investor chooses asset weights to trade off expected return and variance:

$$\max_{\mathbf{w}} \left\{ \mathbf{w}' \mathbf{r} - r_f \mathbf{w}' \mathbf{1} - \frac{a}{2} \mathbf{w}' \boldsymbol{\Sigma}_r \mathbf{w} \right\}. \quad (7)$$

In this paper:

- spreads s_j play the role of expected returns;
- liquidity supplies x_{mj} play the role of portfolio weights;
- covariance of directionality plays the role of return covariance;
- inventory cost replaces risk aversion.

Economic message: Even for one asset, market making is a portfolio problem because the market maker intermediates several distinct order-flow sources.

4 Application to PFOF and order-flow siphoning

4.1 Retail and institutional order flow

The PFOF application has two order sources:

- I -orders, interpreted as institutional orders; and
- R -orders, interpreted as retail orders.

Institutional orders must clear on-exchange. Retail orders may either trade on-exchange or be siphoned off-exchange.

Retail and institutional liquidity demands are linear:

$$\lambda_k(s) = \max\{0, (\zeta - s)\omega_k\}, \quad k \in \{I, R\}. \quad (4)$$

Here ζ is the maximum acceptable half spread and ω_k is the magnitude of type- k liquidity demand.

Interpretation: The demand curve is downward-sloping and has a choke spread ζ .

4.2 Order-flow correlation

Let

$$\sigma_k = \text{sd}(D_k), \quad \rho = \text{corr}(D_I, D_R). \quad (8)$$

The correlation ρ is central.

- If $\rho > 0$, retail and institutional orders tend to be on the same side, raising inventory risk.
- If $\rho < 0$, retail and institutional orders tend to offset each other, lowering inventory risk.

Interpretation: A retail order flow can be valuable for hedging institutional order imbalances when ρ is negative.

4.3 Siphoning variable

Let $\alpha \in [0, 1]$ be the fraction of retail orders siphoned off-exchange.

- $\alpha = 0$: no siphoning; all retail and institutional orders trade on-exchange.
- $\alpha = 1$: all retail orders trade off-exchange; only institutional orders remain on-exchange.

Interpretation: PFOF is modeled as order-flow segmentation: retail orders are routed directly to off-exchange market makers.

4.4 The key threshold Δ

The paper defines

$$\Delta = \sigma_R^2 \omega_R + \rho \sigma_I \sigma_R \omega_I - (\sigma_I^2 \omega_I + \rho \sigma_I \sigma_R \omega_R). \quad (5)$$

Interpretation: Δ compares the marginal inventory consequences of moving retail order flow off-exchange. When $\Delta < 0$, siphoning retail flow is attractive for market makers.

4.5 Proposition 2: PFOF equilibrium

Proposition 2. The equilibrium is characterized by Δ :

- If $\Delta < 0$, there is a unique equilibrium with $\alpha = 1$: all retail orders are siphoned off-exchange.
- If $\Delta > 0$, there is a unique equilibrium with $\alpha = 0$: no siphoning.
- If $\Delta = 0$, there can be multiple equilibria.

Interpretation: PFOF arises endogenously when off-exchange retail flow improves the market maker's inventory portfolio.

4.6 Proposition 3: positive on- and off-exchange volume

Proposition 3. Equilibrium with positive volume both on- and off-exchange exists under realistic parameter restrictions. In particular, if retail demand is smaller than institutional demand and retail order flow is sufficiently useful for inventory management, the with-siphoning equilibrium features positive on-exchange institutional volume and positive off-exchange retail volume.

Interpretation: The model is designed to capture the empirically relevant case: retail orders are not the whole market, but they can be siphoned off-exchange because they help market makers manage risk.

5 Predictions and policy analysis

5.1 Segmentation ban

A segmentation ban forces $\alpha = 0$: all retail and institutional orders must trade on-exchange. Under the ban, there is one on-exchange spread s_b .

Corollary 1. Under a segmentation ban, equilibrium is the same as the baseline model with combined demand:

$$\lambda_1(s) = \lambda_I(s) + \lambda_R(s), \quad \lambda_2(s) = 0. \quad (9)$$

Interpretation: A ban removes the market maker's ability to tailor liquidity supply across retail and institutional order flows.

5.2 Proposition 4: segmentation ban and spreads

Let s_1 be the on-exchange half spread without the ban, s_2 the off-exchange retail half spread, and

$$\bar{s} = \frac{\lambda_I(s_1)s_1 + \lambda_R(s_2)s_2}{\lambda_I(s_1) + \lambda_R(s_2)} \quad (10)$$

be the volume-weighted average half spread.

Proposition 4. In the with-siphoning region,

$$s_2 < s_b, \quad \bar{s} < s_b. \quad (11)$$

Moreover,

$$s_1 < s_b \quad \text{if and only if} \quad \rho < -\frac{M + \gamma\sigma_R^2\omega_R}{\sigma_I\sigma_R\omega_I}. \quad (6)$$

Interpretation: Retail investors are always harmed by a segmentation ban because their spread increases from s_2 to s_b . Institutional investors can also be harmed if retail flow is sufficiently negatively correlated with institutional flow.

5.3 Proposition 5: rise of retail trading

Proposition 5. As the magnitude of retail liquidity demand ω_R increases:

- the off-exchange spread s_2 increases;
- the on-exchange spread s_1 increases if $\rho > 0$ and decreases if $\rho < 0$;
- price improvement $s_1 - s_2$ can rise or fall depending on the correlation structure.

Interpretation: A rise in retail trading can either worsen or improve on-exchange liquidity. The sign depends on whether retail order flow exacerbates or hedges institutional inventory risk.

5.4 Proposition 6: size of the market-making sector

Proposition 6. As the measure of market makers M increases:

- the on-exchange spread s_1 decreases;
- the off-exchange spread s_2 can decrease monotonically or become U-shaped in M .

Interpretation: More competition usually lowers spreads, but inventory hedging can make the effect on off-exchange spreads non-monotone.

5.5 Proposition 7: market-maker profits

Proposition 7. Market makers may earn higher profits when segmentation is banned than under the unregulated PFOF equilibrium, provided the market-making sector is sufficiently large.

Interpretation: PFOF can be a prisoner's dilemma. Each market maker individually wants to siphon retail flow, but collective siphoning changes equilibrium spreads and can lower total market-maker profitability.

5.6 Proposition 8: welfare

Let w be total welfare without a segmentation ban and w_b total welfare under a ban.

Proposition 8. Absent a segmentation ban, the equilibrium maximizes total welfare. Therefore,

$$w > w_b. \tag{12}$$

Interpretation: A ban restricts market makers' portfolio choices and distorts liquidity supply quantities. In this model, that restriction lowers total welfare.

6 Dynamic extension

6.1 Motivation

The appendix extends the model to a dynamic setting in which institutional order flow can trade in a midpoint dark pool before reaching other venues.

Interpretation: This extension addresses the possibility that institutional investors may trade off-exchange for reasons unrelated to PFOF, such as midpoint crossing or dark-pool execution.

6.2 Key logic

The dynamic model has two periods. Outcomes in the first period change the posterior distribution of second-period order flow and therefore change the incentive to siphon retail orders.

- First-period institutional dark-pool executions affect remaining institutional imbalance.
- Market makers update beliefs about order-flow directionality.
- The incentive to siphon retail orders can become stronger or weaker depending on the updated inventory portfolio.

6.3 Proposition 9: dynamic equilibrium

Proposition 9. A full-game equilibrium exists. The equilibrium can involve retail siphoning in the second period, and first-period outcomes affect the second-period siphoning incentive through the posterior distribution of institutional order flow.

Interpretation: The dynamic extension reinforces the core message: market makers segment order flow according to portfolio-level inventory considerations, not only according to the standalone risk of each order type.

7 Tables and figures

7.1 Figure 1: segmentation ban and spreads

Read:

- Figure 1 compares spreads with and without a segmentation ban as order-flow correlation ρ changes.
- The off-exchange retail spread s_2 is below the ban spread s_b .
- The volume-weighted average spread is also below the ban spread.
- The on-exchange institutional spread can be below or above the ban spread depending on ρ .

Economic message: A segmentation ban always worsens retail trading costs and can also worsen institutional trading costs when retail orders hedge institutional imbalances.

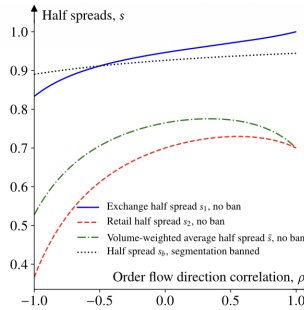


Fig. 1. Bid-ask spreads: the effect of a segmentation ban. This figure shows how various (half) bid-ask spreads change when a segmentation ban is imposed. The order flow direction correlation ρ varies on the horizontal axis. The other parameters are set at $M = 3$, $\gamma = \xi = 1$, $\omega_I = 100$, $\omega_R = 50$, $\sigma_I = 0.5$, $\sigma_R = 0.2$, and $\mu_I = \mu_R = 0$. In particular, $s_1 > s_b$ if and only if $\rho < -\frac{1}{2}$.

7.2 Figure 2: rise of retail trading

Read:

- Panels (a) and (b) compare negative and positive order-flow correlation.
- As retail demand ω_R rises, the off-exchange retail spread s_2 rises.
- The on-exchange spread s_1 falls when retail flow hedges institutional flow and rises when it worsens inventory imbalance.
- Panel (c) gives an example where retail price improvement increases with retail demand.

Economic message: Retail trading growth is not uniformly good or bad for on-exchange liquidity. The inventory-hedging value of retail flow is decisive.

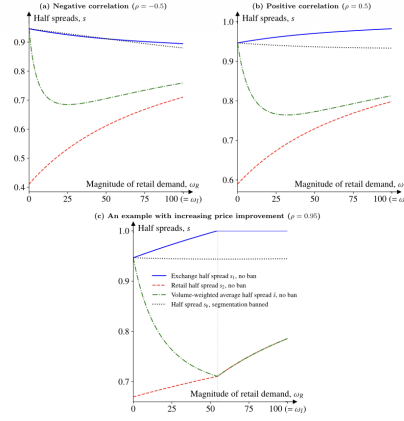


Fig. 2: Bid-ask spreads: the rise of retail trading. This figure shows how various (half) bid-ask spreads change as retail trading demand increases. Panels (a) and (b) illustrate the cases of negative and positive order flow correlations ($\rho = -0.5$ and $\rho = 0.5$, respectively) between retail and institutional order flows. In both Panels (a) and (b), price improvement $s_1 - s_2$ is decreasing in ω_R ; Panel (c) illustrates an example with increasing price improvement (with $\rho = 0.95$). The magnitude of retail demand ω_R varies on the horizontal axes. The other parameters are set at $M = 3$, $\gamma = \zeta = 1$, $\omega_I = 100$, $\sigma_I = 0.5$, $\sigma_R = 0.2$, and $\mu_I = \mu_R = 0$. The vertical dotted line in (c) indicates the upper bound on ω_R implied by (12): when ω_R exceeds that bound, no trading occurs on-exchange.

7.3 Figures 3 and 4: market-maker sector size and profits

Read:

- Figure 3 shows spreads as the measure of market makers M increases.
- The on-exchange spread generally falls with greater competition.
- The off-exchange spread can be U-shaped in M because inventory benefits and competition interact.
- Figure 4 shows market-maker expected profits with and without a segmentation ban.
- For some parameters, market makers are collectively better off under a ban.

Economic message: The model rationalizes why market makers might not uniformly oppose restrictions on PFOF: segmentation can be a prisoner's dilemma.

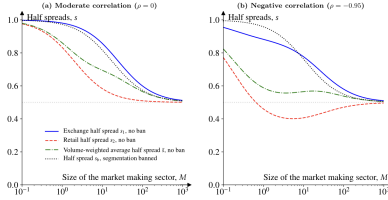


Fig. 3. Bid-ask spreads: the size of the market making sector. This figure shows how various (half) bid-ask spreads change as the market making sector grows. Panels (a) and (b) illustrate the cases of moderate and negative order flow correlations, $\rho = 0$ and $\rho = -0.95$, respectively. The size of the market making sector M varies from 0.1 to 1,000 on the horizontal axis. The other parameters are set at $\gamma = \zeta = 1$, $\omega_I = 100$, $\omega_R = 50$, $\sigma_I = 0.5$, $\sigma_R = 0.2$, and $\mu_I = \mu_R = 0$.

Figure 3: spreads and market-maker sector size

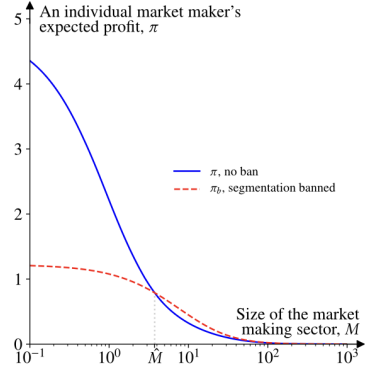


Fig. 4. Market maker profits. This figure shows how the size of the market making sector M affects a market maker's expected profit π with versus without a segmentation ban. The other parameters are set at $\gamma = \zeta = 1$, $\omega_I = 100$, $\omega_R = 50$, $\sigma_I = 0.5$, $\sigma_R = 0.2$, $\rho = -0.3$, and $\mu_I = \mu_R = 0$.

Figure 4: market-maker profits

7.4 Figure 5: welfare

Read:

- Panel (a) varies retail liquidity demand ω_R .
- Panel (b) varies order-flow correlation ρ .
- In both panels, welfare without a segmentation ban exceeds welfare under the ban.

Economic message: The welfare result follows from allowing market makers to choose the liquidity-supply portfolio that minimizes inventory costs while meeting investor demand.

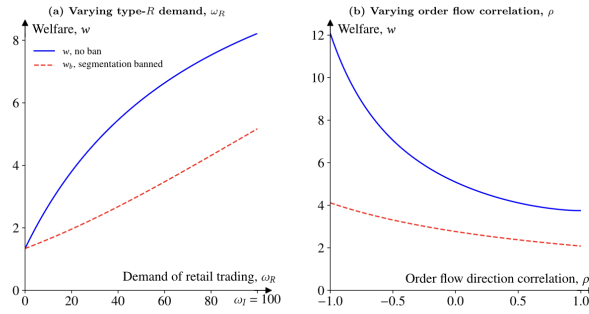


Fig. 5. Welfare. This figure shows how total welfare is affected by the magnitude of retail liquidity demand ω_R in Panel (a) and the order flow correlation ρ in Panel (b). For Panel (a), $\rho = -0.3$; for Panel (b), $\omega_R = 50$. The other parameters are set at $M = 3$, $\gamma = \zeta = 1$, $\omega_I = 100$, $\sigma_I = 0.5$, $\sigma_R = 0.2$, and $\mu_I = \mu_R = 0$.

7.5 Appendix notation table

Read:

- The appendix table summarizes the variables used in the baseline model, the PFOF application, the midpoint dark-pool extension, and the dynamic extension.
- It is useful for tracking the many objects: spreads, liquidity supplies, order-flow directionalities, correlations, and welfare terms.

Economic message: The notation reinforces the paper's structure: a general portfolio-liquidity model first, then a PFOF specialization, then extensions.

<i>Exogenous parameters</i>	
J	number of marketplaces
$\lambda_j(\cdot)$	measure of liquidity-demanding orders on marketplace j
D_j	(random) directionality of liquidity-demanding orders on marketplace j
μ	$\mathbb{E}[(D_1, D_2, \dots, D_J)^T]$
Σ	$\text{var}[(D_1, D_2, \dots, D_J)^T]$
M	measure of market makers
γ	parametrization of market maker inventory costs
<i>Endogenous variables</i>	
s_j	half bid-ask spread of marketplace j
x_{mj}	Poisson intensity of liquidity supply for market maker m on marketplace j
Q_{mj}	(random) volume of market maker m on marketplace j
Z_{mj}	(random) net inventory of market maker m on marketplace j
π_m	expected profit of market maker m
Additional notation used in Sections 3–4	
<i>Exogenous parameters</i>	
$\lambda_k(\cdot)$	measure of type- k liquidity-demanding orders
ζ	maximum acceptable half spread
ω_k	magnitude of type- k liquidity demand
D_k	(random) directionality of type- k liquidity demand
Σ_ω	$\text{var}[(D_I, D_R)^T]$
σ_k	$\text{sd}(D_k)$
ρ	$\text{corr}(D_I, D_R)$
Δ	$(\sigma_R^2 \omega_R + \rho \sigma_I \sigma_R \omega_I) - (\sigma_I^2 \omega_I + \rho \sigma_I \sigma_R \omega_R)$
<i>Endogenous variables</i>	
α	fraction of R -orders routed off-exchange (i.e., to marketplace 2)
$F(\alpha)$	order flow weighting matrix
$\bar{s}$	volume-weighted average half spread
π	market maker profitability
w	total welfare
b	subscript to indicate that segmentation is banned, as in s_b , π_b , and w_b
Additional notation used in Appendix B	
<i>Exogenous parameters</i>	
D_I^γ	(random) I -directionality, before crossing in the midpoint dark pool
Additional notation used in Appendix C	
<i>Exogenous parameters</i>	
t	time period ($t \in \{1, 2\}$); superscripted on other variables as “(t)”
ϕ_k	autocorrelation of type- k order flow
<i>Endogenous variables</i>	
$\pi_m^{(t)}$	a market maker m ’s expected profit, before period t trading
$z_m^{(t)}$	a market maker m ’s inventory after period t trading
$\bar{z}^{(t)}$	market makers’ average inventory after period t trading
$I_m^{(1)}$	$\{z_m^{(1)}, D_I^{(1)}, \bar{z}^{(1)}\}$
$\Delta^{(2)}$	$(\phi_I D_I^{(1)} M \bar{z}^{(1)}) / (\zeta - \frac{\zeta}{2}) - \sigma_I^2 \omega_I$

Appendix B. Extension: adding a midpoint dark pool

8 Short summary

- The paper studies order-flow segmentation from an inventory-management perspective.
- Market makers supply liquidity to several order-flow sources and care about total inventory risk.
- Inventory risk depends on the covariance matrix of order-flow directionalities.
- This makes liquidity supply a portfolio problem.
- Retail order flow can be attractive not because it is uninformed, but because it hedges institutional order flow.
- In the PFOF application, retail orders can be siphoned off-exchange.
- Siphoning arises endogenously when it improves the market maker’s inventory portfolio.
- A segmentation ban always harms retail investors in the model and can also harm institutional investors.
- PFOF can lower market-maker profits through a prisoner’s dilemma.
- A segmentation ban reduces total welfare because it restricts efficient liquidity-supply choices.

- The model makes predictions about retail trading growth, market-maker competition, spreads, price improvement, and welfare.

9 Conclusion

9.1 Core conclusion

Baldauf, Mollner, and Yueshen show that PFOF and order-flow segmentation can arise from inventory management rather than adverse selection. Market makers do not evaluate retail and institutional flow in isolation; they evaluate how each flow fits into an overall liquidity-supply portfolio.

9.2 Economic intuition

If retail orders are negatively correlated with institutional order imbalances, they hedge market-maker inventory risk. This makes retail flow valuable to siphon off-exchange. A regulation that forces all orders back onto the exchange removes that inventory-management tool and can raise spreads for both retail and institutional investors.

9.3 Takeaway

The paper changes the policy interpretation of PFOF. In information-based models, segmentation usually helps retail traders but hurts institutional traders. In this inventory-based model, segmentation can help both groups and improve welfare. The key empirical question is whether retail order flow hedges or worsens market-maker inventory risk.